

Revised Generation Interconnection System Impact Study Report

For Queue Project Q13-0225

West Waynesboro 345 kV
March 20, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0225. The Interconnection Customer proposes a 491.0 MW solar photovoltaic facility interconnecting at the West Waynesboro 345 kV substation in the Pennsylvania area. The request was received on April 07, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 139 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 20 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$87,090,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Carl Bailey - Bon Aqua 138 kV
Contingency: Ft Humbug - Orchard Road 500 kV
Loading: 140.9% (Rating: 393 MVA)
DFAX: 11.7%
Pre-project loading: 126.2%
2. Facility: Bon Aqua - Dry Gap 138 kV
Contingency: Ft Humbug - Orchard Rd 500 kV
Loading: 136.2% (Rating: 401 MVA)
DFAX: 11.7%
Pre-project loading: 121.9%
3. Facility: West Waynesboro 345 kV/138 kV TX
Contingency: West Waynesboro - Pineville 345 kV L
Loading: 126.4% (Rating: 619 MVA)
DFAX: 30.6%
Pre-project loading: 102.1%
4. Facility: W Kelso - Hutchinson Plant 1 138 kV
Contingency: Thurmont - Dry Gap 138 kV
Loading: 116.2% (Rating: 351 MVA)
DFAX: 9.7%
Pre-project loading: 102.6%
5. Facility: W Kelso - Dry Gap 138 kV
Contingency: Thurmot - Dry Gap 138 kV
Loading: 115.1% (Rating: 364 MVA)
DFAX: 9.7%
Pre-project loading: 102.0%

6. Facility: Pineville 345/138kV Transformer
Contingency: WWaynesboro 345/138 kV Autotransformer
Loading: 114.9% (Rating: 581 MVA)
DFAX: 25.7%
Pre-project loading: 93.2%
7. Facility: Thurmont - Catoclin 138 kV
Contingency: W Kelso - Dry Gap 138 kV
Loading: 110.7% (Rating: 224 MVA)
DFAX: 5.2%
Pre-project loading: 99.3%
8. Facility: Catoclin - Altavista Power Station 138 kV
Contingency: W Kelso - Dry Gap 138 kV Line
Loading: 108.2% (Rating: 209 MVA)
DFAX: 5.2%
Pre-project loading: 96.0%
9. Facility: Thurmont - Dry Gp 138 kV
Contingency: W Kelso - Dry Gap 138 kV
Loading: 107.8% (Rating: 370 MVA)
DFAX: 9.2%
Pre-project loading: 95.5%
10. Facility: East Waynesbro - W Waynesboro 138 kV
Contingency: Chambersburg 345 kV/138 kV Transformer
Loading: 105.9% (Rating: 356 MVA)
DFAX: 16.8%
Pre-project loading: 82.7%
11. Facility: Dundee - Metro Ctr 138 kV Line
Contingency: Athalmer Tap - Cameron 138 kV
Loading: 105.1% (Rating: 264 MVA)
DFAX: 8.4%
Pre-project loading: 89.4%
12. Facility: Athalmer Tap - Cameron 138 kV
Contingency: Dundee - Metro Center 138 kV
Loading: 103.9% (Rating: 360 MVA)
DFAX: 11.2%
Pre-project loading: 88.6%
13. Facility: Innerarity - Pineville 138 kV
Contingency: Pineville - E Hagerstown Substation 138 kV
Loading: 103.2% (Rating: 386 MVA)
DFAX: 9.5%
Pre-project loading: 91.1%
14. Facility: Hollister - El Sobrante 138 kV
Contingency: Pineville - E Hagerstown 138 kV
Loading: 102.1% (Rating: 350 MVA)
DFAX: 9.5%
Pre-project loading: 88.8%
15. Facility: Messick - Pluess 138 kV
Contingency: Metro Ctr - Dundee 138 kV
Loading: 102.0% (Rating: 336 MVA)
DFAX: 11.2%
Pre-project loading: 85.6%
16. Facility: Five Channels Substation - Messick 138 kV Line
Contingency: Dundee - Metro Ctr 138 kV

Loading: 102.0% (Rating: 309 MVA)

DFAX: 11.2%

Pre-project loading: 84.2%

17. Facility: Athalmer Tap - East Hagerstown 138 kV L

Contingency: Innerarity - Pineville 138 kV

Loading: 101.7% (Rating: 455 MVA)

DFAX: 12.1%

Pre-project loading: 88.7%

18. Facility: Bellsfork Substation - Maryland Solar 138 kV

Contingency: Pineville Substation - E Hagerstown 138 kV

Loading: 100.8% (Rating: 383 MVA)

DFAX: 9.5%

Pre-project loading: 88.7%

19. Facility: E Hagerstown - Pineville Substation 138 kV

Contingency: Innerarity Substation - Pineville 138 kV

Loading: 100.6% (Rating: 492 MVA)

DFAX: 12.1%

Pre-project loading: 88.6%

20. Facility: Hutchinson Plant 1 - Watford City 138 kV

Contingency: Dry Gap - Thurmont 138 kV

Loading: 100.3% (Rating: 323 MVA)

DFAX: 6.3%

Pre-project loading: 90.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Carl Bailey - Bn Aqua 138 kV	Rebuild line with higher-capacity conductor	\$4,511,000	100% (\$4,511,000)
Bon Aqua - Dry Gap 138 kV L	Rebuild line with higher-capacity conductor	\$7,165,000	100% (\$7,165,000)
West Waynesboro 345-138 kV Autotransformer	Replace transformer with higher-rated unit	\$24,901,000	94% (\$23,430,000)
Hutchinson Plant 1 - W Kelso 138 kV Line	Rebuild line with higher-capacity conductor	\$5,727,000	100% (\$5,727,000)
West Kelso - Dry Gap 138 kV	Rebuild line with higher-capacity conductor	\$2,444,000	100% (\$2,444,000)
Pineville 345/138kV Transformer	Replace transformer with higher-rated unit	\$21,225,000	82% (\$17,402,000)
Thurmont - Catoctin 138 kV line	Rebuild line with higher-capacity conductor	\$5,769,000	100% (\$5,769,000)
Catoctin - Altavista Power Station 138 kV	Reconductor line	\$2,241,000	100% (\$2,241,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Thurmont - Dry Gp 138 kV	Reconductor line	\$1,515,000	100% (\$1,515,000)
East Waynesbro - W Waynesboro 138 kV	Reconductor line	\$2,056,000	100% (\$2,056,000)
Dundee - Metro Ctr 138 kV	Reconductor line	\$8,392,000	76% (\$6,392,000)
Athalmer Tap Substation - Cameron 138 kV line	Reconductor line	\$1,246,000	77% (\$956,000)
Innerarity - Pieville 138 kV	Reconductor line	\$863,000	76% (\$660,000)
Holister - El Sobrante Substation 138 kV	Reconductor line	\$1,212,000	76% (\$926,000)
Messick - Pluess 138 kV	Reconductor line	\$2,967,000	77% (\$2,276,000)
Five Channels Substation - Messick 138 kV Line	Reconductor line	\$743,000	77% (\$570,000)
Athalmer Tap - East Hagerstown 138 kV L	Reconductor line	\$2,071,000	42% (\$866,000)
Bellsfork Substation - Maryland Solar 138 kV	Reconductor line	\$1,340,000	76% (\$1,024,000)
Pineville - East Hagerstown 138 kV	Reconductor line	\$1,299,000	42% (\$543,000)
Hutchinson Plant 1 - Watford City 138 kV	Reconductor line	\$617,000	100% (\$617,000)
	Total allocated to project		\$87,090,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.